
Portfolio
Computer Vision:
– Sheet 3 –

Portfolio 3 includes one integrated exercise.

Your goal is to apply your knowledge of Explainable AI and Edge AI in two integrated, real-world scenarios. Each part is worth 15 points. You must complete both parts to achieve the full score.

Your submission should include:

- Implementation code (as well-structured Python notebooks or scripts following best coding practices)
- Generated outputs and visualizations (e.g., reconstructions, interpolations, latent traversals)
- A short presentation covering:
 - Your approach and design decisions
 - Training pipeline and experimental setup
 - Key evaluation results and comparisons
 - Reflections on trade-offs and insights

This is a group assignment. Each team should consist of 3 members.

Important: This time, each group will be examined individually in a technical due diligence session (see details below).

Dataset

You will use the CelebA dataset (Aligned + Cropped version). CelebA contains over 200,000 celebrity face images with attribute annotations.

Download: <https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html>

Direct Link:

<https://drive.google.com/drive/folders/0B7EVK8r0v71pWEZsZE9oNnFzTm8?resourcekey=0-5BR16BdXnb8hVj6CNHKzLg&usp=sharing>

Part 1: Practice (15 Points)

Implement and analyze a Variational Autoencoder on CelebA. Your practical work should include:

Task 1: Baseline VAE Implementation

- Build and train a VAE on CelebA
- Document your architecture, latent dimension choice, and training setup
- Report training curves

Task 2: Latent Space Exploration

- Visualize reconstructions for at least 5 test images
- Perform latent interpolations between pairs of images (minimum 3 sequences)
- Conduct latent traversals by varying individual latent dimensions and show their effect on generated images

Task 3: β -VAE Variant

- Train a β -VAE with at least two different β values
- Compare reconstruction quality and latent disentanglement qualitatively and discuss trade-offs

Part 2: Theory (15 Points)

Answer the following questions in a concise, well-structured manner (include formulas and diagrams where you think it could be helpful):

1. Latent Dimension Choice

- a. How does the size of the latent space affect reconstruction quality and generative diversity?
- b. What practical signs indicate that your latent dimension is too small or too large?

2. KL vs. Reconstruction Trade-off

- a. During training, what does it mean if the KL term dominates the loss?
- b. How would you adjust your training strategy to address this?

3. Data Preprocessing Impact

- a. Why is image alignment and cropping critical for CelebA when training VAEs?
- b. What could happen if you skip normalization or resizing?

4. Training Stability

What symptoms indicate posterior collapse during training?

5. Latent Space Manipulation

How can you estimate directions in latent space that correspond to semantic attributes (e.g., "Smiling")?

Technical Due Diligence (Shark Tank Format)

This portfolio will conclude with an individual examination session for each group:

- Ownership: Each team member must clearly indicate in the report and notebooks which components they were primarily responsible for (e.g., data preprocessing, model design, latent analysis, which questions in the theory part, etc.)
- Live Q&A: In a short session after your presentation, each member will answer targeted questions about their own work and core concepts of the project
- Expectation: You must be able to justify design decisions, explain key results, and respond to conceptual questions. Simply “I was just the one who was running the code” is not sufficient
- Evaluation: Individual performance in the Q&A will influence the amount of points in this portfolio (up to $\pm 10\%$)

Note: Hardware Constraints

CelebA is a large and resource-intensive dataset. **You are allowed** to apply reasonable optimizations (e.g. downscaling images, etc.) to make training feasible on your hardware.

However, these optimizations **must not compromise or skip** any of the mandatory analyses and tasks outlined in the assignment. All required experiments, visualizations, and comparisons must still be completed as specified.